Interest rate risk

Introduction

- Is the risk that arises from the change in the rate of interest on borrowed or invested money.
- Interest rate exposure = Amount of loan or investment x change in interest rate.
- The exposure is the gain/loss associated with change in interest rate.
- A small change in interest rate on borrowed funds can cause a significant change in the reported profits.

Spot and forward interest rate.

- A spot interest rate ("spot rate") is a rate of interest on a security that makes a single payment at a future point in time.
- Each cash flow should be discounted at a unique interest rate that is appropriate for the time period in which the cash flow will be received i.e. the spot rate.
- A forward interest rate is the rate that applies on investments that will start in future and for a specific future period.
- Spot interest rates directly affects forward interest rates.

Pricing of securities.

- The value of any financial security is the sum of the present values of all future cash flows associated with a particular financial asset.
- For debt instruments the discount rate used is the current market interest rate.
- Each cash flow should be discounted at a unique interest rate that is appropriate for the time period in which the cash flow will be received i.e. the spot rate.
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Spot and forward interest rate symbols and meaning.

- **Spot rates can be denoted as:**
- ${}_0S_1$ = 1-year spot rate
- ${}_0S_2$ = 2-year spot rate
- ${}_0S_3$ = 3-year spot rate
- **Forward interest rates can be denoted as:**
- ${}_1f_2$ = 1-year forward rate that begins in year 1
- ${}_1f_3$ = 2-year forward rate that begins in year 1
- ${}_2f_5$ = 3-year forward rate that begins in year 2

Spot and forward interest rate linked.

- Spot rates and forwards rates can be calculated using spot rates or a combination of spot and forward rates.
- For example, a one-year forward rate can be calculated using the 2-year spot rate and 1-year spot rate as follows;
- $(1+ {}_0S_2)^2 = (1 + {}_0S_1)^1 (1 + {}_1f_2)^1$
- 3-year spot rates can used in the following ways.
- $(1+ {}_0S_3)^3 = (1 + {}_0S_2)^2 (1 + {}_2f_3)^1$ OR
- $(1+ {}_0S_3)^3 = (1 + {}_0S_1)^1 (1 + {}_1f_3)^2$ OR
- $(1+ {}_0S_3)^3 = (1 + {}_0S_1)^1 (1 + {}_1f_2)^1 (1 + {}_2f_3)^1$

Spot and forward interest rate linked.

- For example, a one-year forward rate can be calculated using the 2-year spot rate and 1-year spot rate as follows;
- $(1+ {}_0S_2)^2 = (1 + {}_0S_1)^1 (1 + {}_1f_2)^1$
- Tasks
- Determine the 1-year forward rate that begins in year 1 if the 2-year spot rate is 6.25% and the 1-year spot rate is 5%
- $(1+ 0.0625)^2 = (1 + 0.05)^1 (1 + {}_1f_2)^1$
- ${}_1f_2 = \frac{(1.0625)^2}{(1.05)^1} - 1 = \underline{7.51\%}$

Activity 1.

- **The following are the prices of a one-year, two-year, three-year, and four-year government zero coupon bonds respectively; K86.96, K77.64, K71.89, K65.35.**
- **<u>Required:</u>**
- **(i) Calculate the set of spot interest rates that can be obtained from these bonds.**
- **(ii) Using the spot interest rates calculated above, calculate the two-year forward rate starting in two years' time and the one-year forward rate starting in three years' time. . (a)= $_2f_4$ (b) = $_3f_4$**

Solution to Activity 1.

- **Value of zero-coupon bond (Po) = $\frac{\text{Face value}}{(1 + {}_0S_n)^n}$**

- **Where :**

- **${}_0S_n$ = Spot rate for period n**

- **n = Time to maturity**

Solution to Activity 1.

- Value of zero-coupon bond (Po) = $\frac{\text{Face value}}{(1 + {}_0S_n)^n}$
- **Where :**
- ${}_0S_n$ = Spot rate for period n
- n = Time to maturity
- One-year zero coupon bond price at K86.96; n = 1
- K86.96 = $\frac{\text{K100}}{(1 + {}_0S_1)^1}$ K86.96 *$(1 + {}_0S_1)^1$ = K100
- ${}_0S_1 = \frac{100}{86.96} - 1 = 15\%$

Solution to Activity 1.

- Value of zero-coupon bond (Po) = $\dfrac{\text{Face value}}{(1 + {}_0S_n)^n}$
- Two-year zero-coupon bond price at K77.64; n = 2
- K77.64 = $\dfrac{\text{K100}}{(1 + {}_0S_2)^2}$
- K77.64 *$(1 + {}_0S_2)^2$ = K100
- ${}_0S_2 = \sqrt{\left(\dfrac{100}{77.64}\right)} - 1 = 13.5\%$

Solution to Activity 1.

- Value of zero-coupon bond (Po) = $\dfrac{\text{Face value}}{(1 + {}_0S_n)^n}$
- Three-year zero-coupon bond price at K71.89; n = 3
- K71.89 = $\dfrac{\text{K100}}{(1 + {}_0S_3)^3}$
- K71.89 *$(1 + {}_0S_3)^3$ = K100
- ${}_0S_3 = \sqrt[3]{\left(\dfrac{100}{71.89}\right)} - 1 = 11.6\%$

Solution to Activity 1.

- Value of zero-coupon bond (Po) = $\frac{\text{Face value}}{(1 + {}_0S_n)^n}$
- Four-year zero-coupon bond price at K65.35; n = 4
- K65.35 = $\frac{\text{K100}}{(1 + {}_0S_4)^4}$
- K65.35 *$(1 + {}_0S_4)^4$ = K100
- ${}_0S_4 = \sqrt[4]{\left(\frac{100}{65.35}\right)} - 1 = 11.2\%$

Interest rate and bond duration

- **Duration is a measure of the number of periods into the future where a bond's value, on average, is generated.**
- **The greater the duration of a bond, the further into the future its average value is generated, and the more its value will react to changes in interest rates.**
- **Duration shows the extent to which the bond's value will change for every 1% change in the market interest rate.**
- **It shows the *sensitivity* of the bond price to changes in interest rates.**

Calculating Bond duration

- **Duration is calculated as = $\frac{\Sigma \text{ (PV of CF x t)}}{\text{Bond price}}$**

  Where: t = specific time when cash flow is generated

  PV of CF = present value of specific cash flows

- **Bond duration is measured as a period.**
- **It shows the payback period of the bond's cash flows.**

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Activity 2

MBS has issued 10% K1,000, 4-year bonds . The bonds are redeemable at their nominal value while interest is paid annually. The spot rates over the next 4 years are: ${}_0S_1$ =4%, ${}_0S_2$ = 4.5% ; ${}_0S_3$ =5%; ${}_0S_4$ = 6%

<u>Required:</u>

a) **Calculate the bond's current market price and bond duration**

b) **Determine the percentage change in bond's price if the spot interest rate declined by 50 basis points for years 1 and year 4 only.**

- **Hint: 100 basis points = 1%; 1 basis point = 0.01%**

Solution to practice question

Spot rate	Year (t)	Cash flow K	Discount factor at spot rate	Present Value (K)	PV *t
4%	1	100	0.962	96.20	96.20
4.5%	2	100	0.916	91.60	183.20
5.0%	3	100	0.864	86.40	259.20
6.0%	4	1100	0.792	871.20	3,484.80
			Sum of PVs	*1,145.40*	*4,023.4*

Solution to practice q.

- **Duration is** $= \dfrac{\Sigma \text{ PV of CF x t}}{\text{Bond price}}$
- $= 4{,}023.40/1{,}145.40$
- $= 3.51$ **years**
- **This means that it will take approximately 3.51 years for an investor in this instrument to recoup their investment.**
- **In addition, the result indicates that the price/value of this bond will change by 3.51 for every 1% change in the market interest rate.**

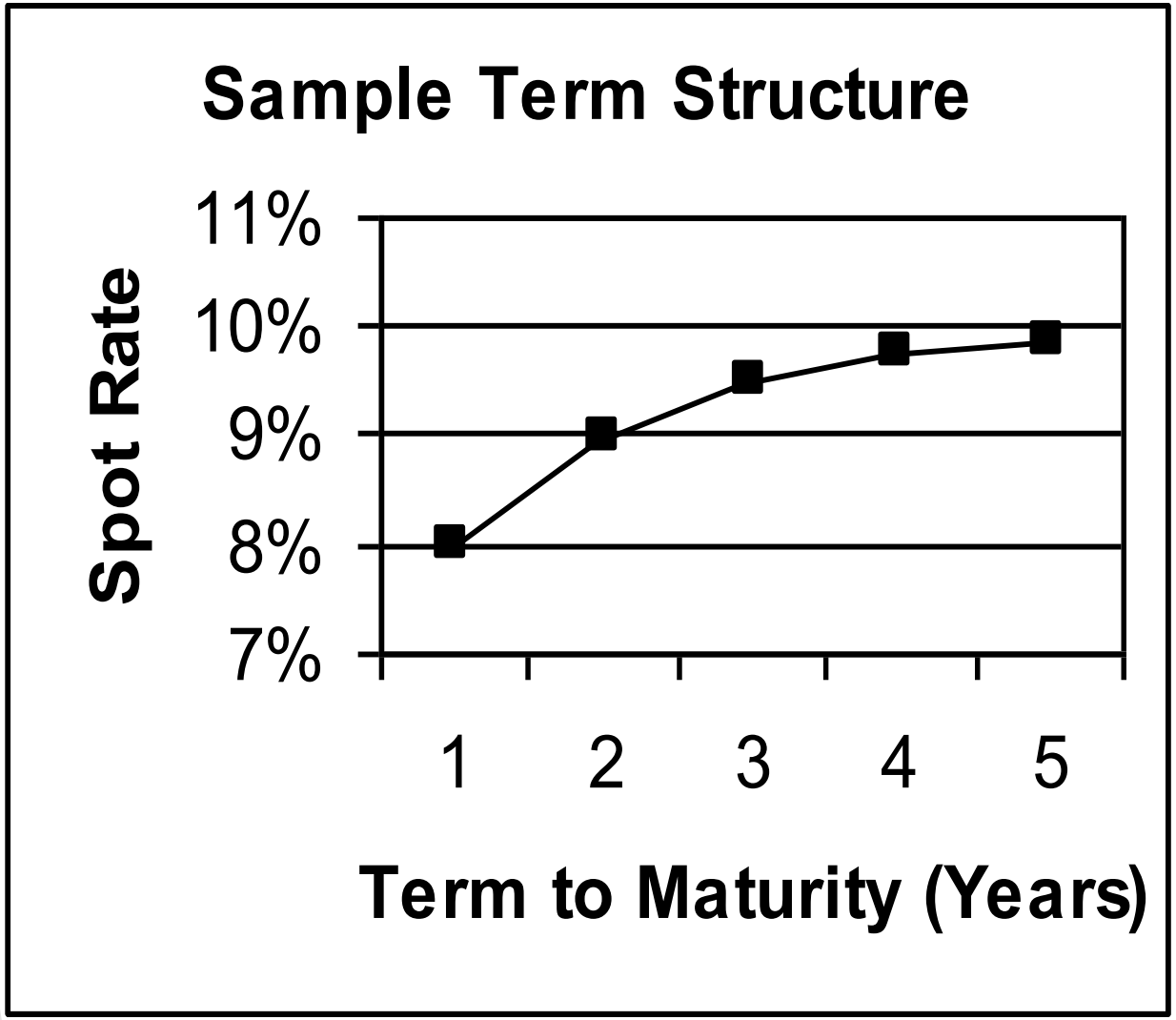
Activity 3

- **Calculate the bond duration of the following bonds given that market interest rate is 6% per year.**
- **Bond A: 10% K1,000 bond redeemable at par in 5 years times.**
- **Bond X: 12% K1,000 redeemable at a 2% premium in 3 years time.**
- **Bond Z: K1,000 zero coupon bond redeemable at par in 5 years time.**

Term structure of interest rate.

- Term structure of interest rates shows the relationship between interest rates and time.
- This can be downward sloping, upward sloping or even flat.
- This is graphically shown as a yield curve.
- Implied forward interest rates, and expected future interest rates, all depends on the yield curve.

The term structure of interest rates is the relation between different interest rates for different term-to-maturity loans.



If we observe $r_1 = 8\%$, $r_2 = 9\%$, $r_3 = 9.5\%$, $r_4 = 9.75\%$ and $r_5 = 9.875\%$

Then the current term structure of interest rates is represented by plotting these "spot rates" against their terms-to-maturity.

The curve plotted through the above points is also called the "yield curve"

Traditional term structure theories

- **Expectation theory**
- **Liquidity Preference**
- **Market segmentation theory (Preferred habitat)**
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Expectation theory

- The expected return on a security of whatever maturity is the same for the same holding period.
- The shape of the yield curve is determined by expected short term interest rates and the relationship with passage of time.
- Arbitrage actions would cause prices of bonds to change until equilibrium.
- Forward rates can be determined using the expectation theory i.e. spot rates and other known forward rates.

Liquidity Preference.

- Argues that investors require a risk premium as reward for holding longer maturity investments (instruments).
- The longer the maturity period, the higher is the reward(premium) needed to motivate investors to put their money away for longer periods.
- The shape of the yield curve is thus upward sloping.
- The implied forward interest rate maybe a biased predictor of future spot rate for that maturity.

Market Segmentation theory (Preferred habitat).

- **Argues that *market participants have a preferred maturity range* or habitat in which they like to borrow or lend.**
- **Future interest rates are influenced by demand and supply and expectation of future interest rates.**
- **When interest rates are rising, borrower will prefer longer maturity rates(term) while investors will prefer short maturity rates and vice versa.**

Internal hedging.

- ***Matching:***
- **This is an attempt to have the same interest rates on both investments and sources of finance.**
- **This implies that assets and liabilities will have the same interest rate.**
- **For example, a financial institution can match interest received from lenders with interest given to depositors**

Internal hedging.

- ***Smoothing:***
- **Maintenance of a balance between fixed rates and floating rate borrowing i.e. not having too much fixed rate debts or too much of floating rate debt.**

External hedging.

- The external hedging methods includes the following:
- 1. Forward Rate Agreements (FRAs)
- 2. Interest Rate Guarantee (IRGs)
- 3. Interest Rate Futures (IRFs)
- 4. Interest Rate Swaps (IRS)

Forward Rate Agreements (FRAs)

- This involves fixing the interest rate today for future short-term borrowing and depositing.
- An FRA is an over-the-counter contract that is tailored to the needs of the parties i.e., maturity dates and periods can be tailored to the company's requirements.
- Interest rates are quoted on a bid/offer basis, so a quote of 6.50 – 6.40 means that a company can fix a borrowing rate at 6.5% and fix a deposit rate at 6.40%.
- The rate agreed is based on a notional amount starting on a date in future and lasting for a specific period e.g., 3 v 9 means that the FRA starts in 3 months and lasts for 6 months

Activity 4

- **MBS Co will shortly be making a short-term investment and wants to borrow K8m for 3 months, starting in 3 months' time. However, interest rates are currently volatile, and the firm is worried about adverse movements in these rates before it takes out the loan. LIBOR (the base rate) is currently 3.5% and MBS has been offered a 3 v 6 FRA for 3.75%. MBS can currently borrow at approximately 1% above the base rate.**
- **<u>REQUIRED:</u>**
- **Advise MBS of the likely outcome of the FRA if the base rate in 3 months' time is 4%.**

Solution to activity 4

FRA : 3 V 6, agreed rate = 3.75%, Borrowing rate = 5%, Notional amount = K8m

At 5% interest, the bank will pay MBS because interest has risen.	**K**
FRA receipt = K8m *(5% - 3.75%)*3/12	**25,000**
Payment on the loan =5%*K8000000*3/12	**(100,000)**
Net payment on the loan	**(75,000)**
Effective interest rate on the loan = (75000/8000000)*12/3	**3.75%**

Interest rate guarantee (IRG)

- **This involves *entering into an agreement* that fixes the rate at which one can borrow or deposit in future like in FRA.**
- **However, IRG provides the right but not an obligation.**
- **Therefore, if interest rates are not favorable, the option holder can abandon the contract.**
- **A premium is paid to the IRG writer for accepting the downside risk. IRG can exchange traded or tailor-made.**

Interest rate futures (STIR)

- **This is an *exchange traded instrument* that involves fixing the interest rates like FRA except that the terms, amounts and periods are standardized.**
- **A gain on futures covers the loss or increase in interest expense on company debt**

Interest rate swaps

- This is an exchange between two parties of interest obligations or receipts in the same currency on an agreed amount of notional principal for an agreed period.
- Interest rate swaps provide a flexible method for firms to change interest rate profiles of their underlying loans or investments.
- A company which has debt at a fixed rate of interest can make a swap so that it ends up paying interest at a variable rate and vice versa.